Task Three: Telecom Customer Churn Prediction

Datheon Machine Learning & Data Science Internship | Technical Report

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1. Executive Summary

This technical report details the end-to-end development, pipeline engineering, evaluation, and operational insights for Task Three of the Datheon ML & DS Internship. The goal was to build a machine learning pipeline to predict telecom customer churn using subscription records, preprocess categorical and numerical features without data leakage, handle class imbalance, and achieve a test set ROC-AUC score exceeding the required 80% benchmark.

> **TASK BENCHMARK ACHIEVED**
> The trained XGBoost classifier achieved a 5-Fold Cross-Validation ROC-AUC score of 0.8485 ± 0.0109 and a Test Set ROC-AUC score of 0.8447 (0.836 in visual evaluation), cleanly surpassing the required >80% threshold. The Random Forest pipeline achieved 0.8411 ± 0.0083 CV ROC-AUC and 0.8363 Test ROC-AUC.

2. Problem Statement & Technical Requirements

Customer churn poses a significant financial risk in the telecommunications sector. To address this, the project followed a structured ML development workflow:

- **1. Data Ingestion & Inspection:** Load the Telco Customer Churn dataset (7,043 rows, 21 columns) and verify data types and missing value counts.
- **2. Data Cleaning & Cleaning:** Convert 'TotalCharges' from string to numeric type, handle potential missing values, and map the target variable ('Churn') to binary integers.
- **3. Leakage-Free Preprocessing:** Partition numeric and categorical features and construct scikit-learn ColumnTransformers with SimpleImputers, StandardScaler, and OneHotEncoder.
- **4. Class Imbalance Mitigation:** Compare synthetic over-sampling via SMOTE inside imblearn pipelines against cost-sensitive reweighting using scale_pos_weight in XGBoost.
- **5. Cross-Validation & Evaluation:** Validate model generalization using 5-Fold Stratified Cross-Validation, evaluate test set ROC-AUC (>80%), and extract top 10 feature importances.

3. Dataset Analysis & Data Preprocessing

3.1 Data Cleaning & Type Conversion

The original dataset contains 7,043 customer records across 21 attributes. During initial inspection, the 'TotalCharges' feature was identified as a string object due to whitespace characters present for new customers with zero tenure. The field was forcibly coerced to numeric using pd.to_numeric(..., errors='coerce'). After conversion, all 7,043 entries were verified. Non-predictive fields such as 'customerID' were removed from the feature matrix.

The target variable 'Churn' was converted from 'Yes'/'No' labels to binary integers (1 / 0). Target class distribution revealed 1,869 churn instances (26.5%) and 5,174 non-churn instances (73.5%), establishing an imbalanced class ratio of approximately 2.77:1.

3.2 Pipeline Architecture & Data Leakage Prevention

To ensure statistical validity and strictly prevent data leakage during cross-validation, imputation, scaling, and encoding were modularized into a ColumnTransformer pipeline:

- **Numerical Stream (4 features):** StandardScaler combined with SimpleImputer(strategy='median').
- **Categorical Stream (15 features):** OneHotEncoder(sparse_output=False, handle_unknown='ignore') combined with SimpleImputer(strategy='most_frequent').
- **Expanded Feature Space:** The OneHotEncoder expanded the 15 categorical features into 41 binary columns, creating a total transformed feature matrix of 45 features.

4. Model Development & Class Imbalance Strategies

4.1 Random Forest + SMOTE Pipeline

To balance target distributions for Random Forest, Synthetic Minority Over-sampling Technique (SMOTE) was embedded directly inside an imblearn.pipeline.Pipeline. This guarantees that synthetic sampling occurs strictly within each cross-validation training fold, preventing test fold contamination. The classifier was instantiated with n_estimators=200, max_depth=10, and random_state=42.

4.2 XGBoost + Cost-Sensitive Reweighting (`scale_pos_weight`)

As an alternative to synthetic sample generation, XGBoost was configured with cost-sensitive loss weighting. The scale_pos_weight parameter was dynamically computed as:

$$\text{scale_pos_weight} = (\text{Count of Negative Class}) / (\text{Count of Positive Class}) \approx 2.77$$

The XGBClassifier was configured with n_estimators=150, max_depth=4, learning_rate=0.05, eval_metric='logloss', and random_state=42.

5. Performance Evaluation & Validation Results

Models were evaluated using 5-Fold Stratified Cross-Validation on the 80% training partition (5,634 samples), followed by hold-out evaluation on the 20% test partition (1,409 samples).

Model Pipeline	Imbalance Method	5-Fold CV ROC-AUC	Test Set ROC-AUC	Requirement (>80%)
Random Forest Classifier	SMOTE (Over-sampling)	0.8411 ± 0.0083	0.8363	**PASSED** ☑
XGBoost Classifier	scale_pos_weight	0.8485 ± 0.0109	**0.8447**	**PASSED** ☑

6. Feature Importances & Strategic Business Insights

Extracting feature importances from the fitted XGBoost model revealed the top 10 most influential factors driving customer churn:

- **1. Month-to-Month Contracts (Contract_Month-to-month):** By far the most dominant feature (importance > 0.52). Customers without long-term commitments churn at significantly higher rates.
- **2. Fiber Optic Internet Service (InternetService_Fiber optic):** The second most influential factor (~0.08 importance). Fiber optic users show elevated churn relative to DSL and non-internet users, indicating potential price dissatisfaction or service issues.
- **3. Absence of Tech Support & Security:** Features like 'OnlineSecurity_No' and 'TechSupport_No' rank in the top 4 importances. Customers lacking add-on support services exhibit higher vulnerability to churn.
- **4. Payment Method & Contract Terms:** Electronic check payment methods ('PaymentMethod_Electronic check') and multi-year contract terms ('Contract_One year', 'Contract_Two year') serve as critical positive/negative retention indicators.
- **5. Tenure (Account Lifespan):** Customer tenure strongly correlates inversely with churn probability; short-tenure subscribers represent the highest attrition risk.

7. Conclusions & Deployment Recommendations

- **Model Recommendation:** XGBoost with scale_pos_weight is recommended for production deployment due to superior ROC-AUC performance (0.8485 CV / 0.8447 Test) and zero synthetic sample generation overhead.
- **Retention Strategy:** Target month-to-month subscribers with incentives to switch to annual plans, and investigate quality-of-service/pricing feedback specifically from Fiber Optic users.
- **Production Integration:** The entire preprocessing and modeling pipeline can be serialized using joblib for real-time batch and REST API scoring.